Comprehensive Guide to Kubernetes Architecture and Deployments

Kubernetes is an open-source container orchestration platform designed to automate deploying, scaling, and operating application containers. At its core, a Kubernetes cluster consists of control plane components and worker nodes. The control plane runs the API Server (kube-apiserver), which validates and configures data for objects such as pods, services, and replication controllers. The etcd distributed key-value store provides reliable state storage. The kube-scheduler observes newly created Pods and assigns them to nodes based on resource requirements, constraints, and affinity specifications.

Worker nodes host the components responsible for running containerized workloads: the kubelet agent, kube-proxy network proxy, and the container runtime (such as containerd). Pods are the smallest deployable computing units. Higher-level abstractions like Deployments declare desired state, enabling declarative zero-downtime rolling updates. Services provide stable IP addresses and DNS names to route traffic to dynamic pod endpoints. Ingress controllers manage external HTTP/S access into the cluster, providing load balancing, SSL/TLS termination, and name-based virtual hosting.